IT Workshop II (Cloud Computing): Cloud Security, Big Data Analytics

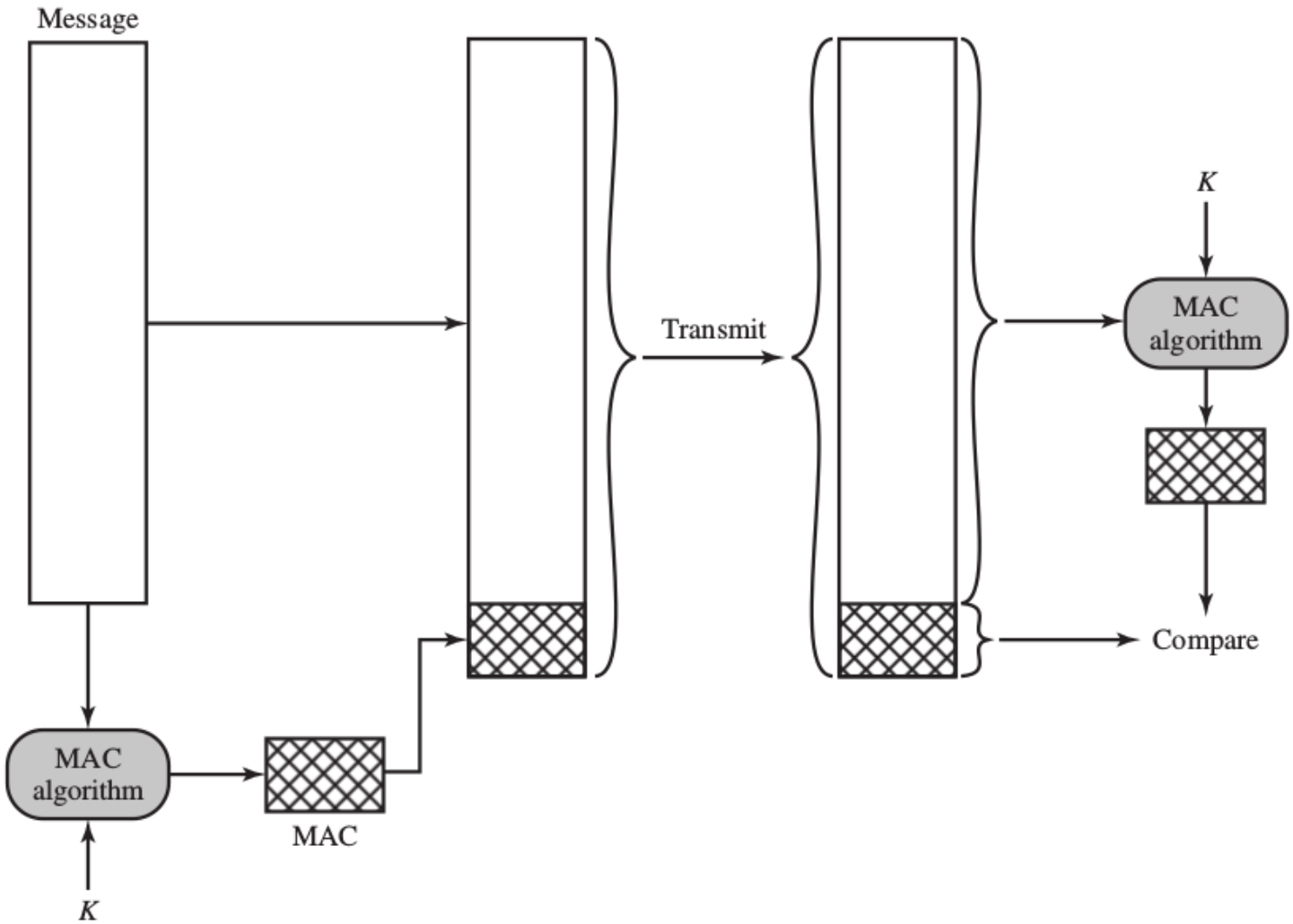
October 26, 2026

Message Authentication Code

1. This authentication technique involves the use of a **secret key** to generate a **small block of data**, known as a **message authentication code**, that is appended to the message.

2. When **A** has a message to send to **B**, it calculates the message authentication code as a complex function of the message and the key:
   $MAC_M = \mathrm{F}(K_{AB}, \mathrm{M})$.

Message Authentication Code



Message Authentication Code

As messages may be of any size and the message authentication code is a small fixed size, there must theoretically be many messages that result in the same MAC.

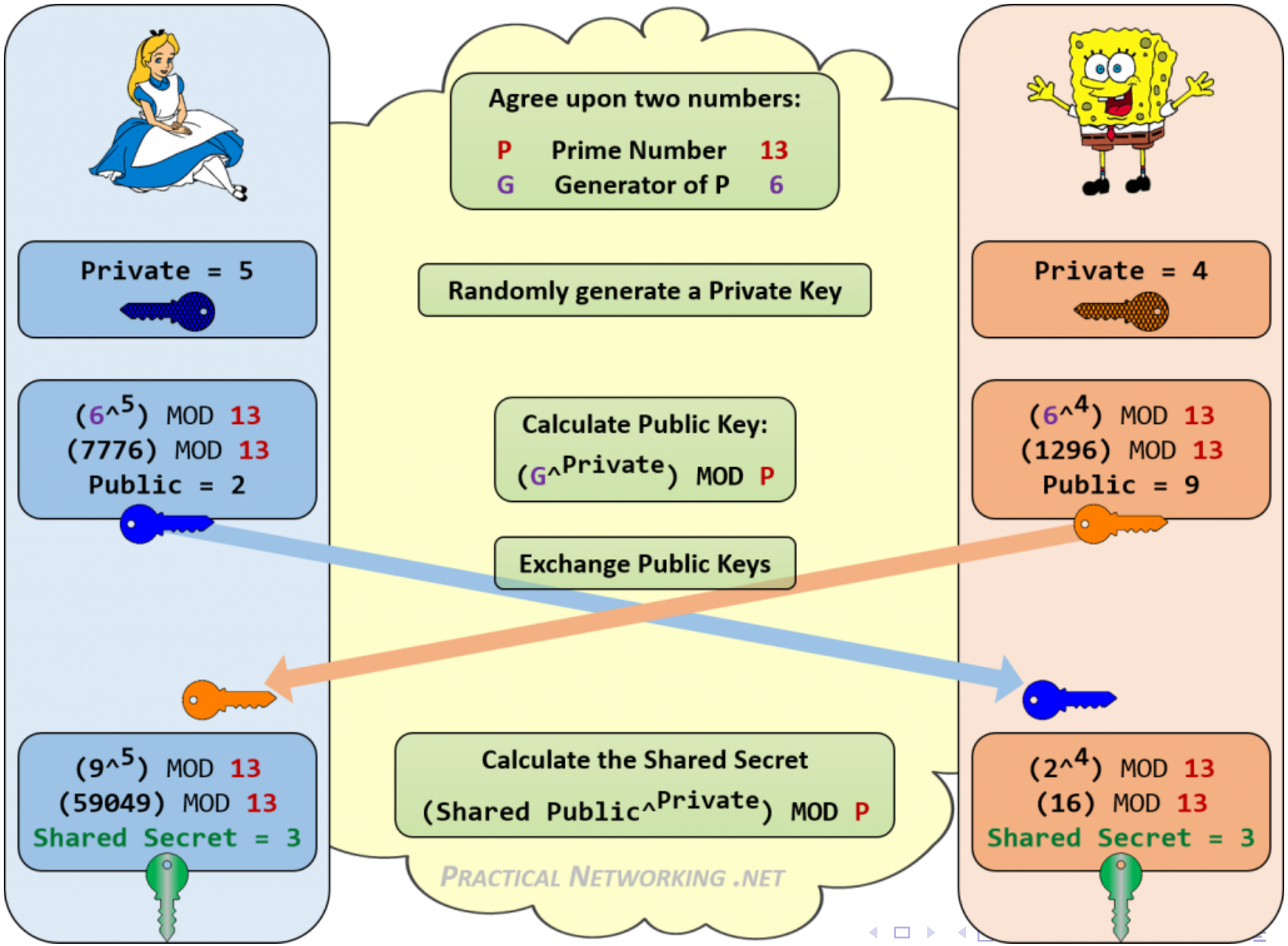
However, it should be **infeasible in practice** to find pairs of such messages with the same MAC. This is known as **collision resistance**.

Message Authentication Code

MAC Ensures:

- The receiver is assured that the message has not been altered. Because the attacker is assumed not to know the secret key, the attacker cannot alter the MAC to correspond to the alterations in the message.
- The receiver is assured that the message is from the alleged sender.(Because of the assumption that the key is secret)
- If the message includes a sequence number, then the receiver can be assured of the proper sequence.

Diffie-Hellman Key Exchange



Diffie-Hellman Key Exchange

- Diffie-Hellman provides no authentication of the two communicating partners. There are variations to Diffie-Hellman that overcome this problem.

Big Data

- Big data is defined as collections of data sets whose volume, velocity in terms of time variation, or variety is so large that it is difficult to store, manage, process and analyze the data using traditional databases and data processing tools.
- Characteristics of big data:
  - **Volume (Scale):** Though there is no fixed threshold for the volume of data to be considered as big data, however, typically, the term big data is used for massive scale data that is difficult to store, manage and process using traditional databases and data processing architectures.
  - **Velocity (Speed):** Velocity is another important characteristic of big data and the primary reason for exponential growth of data.

Big Data

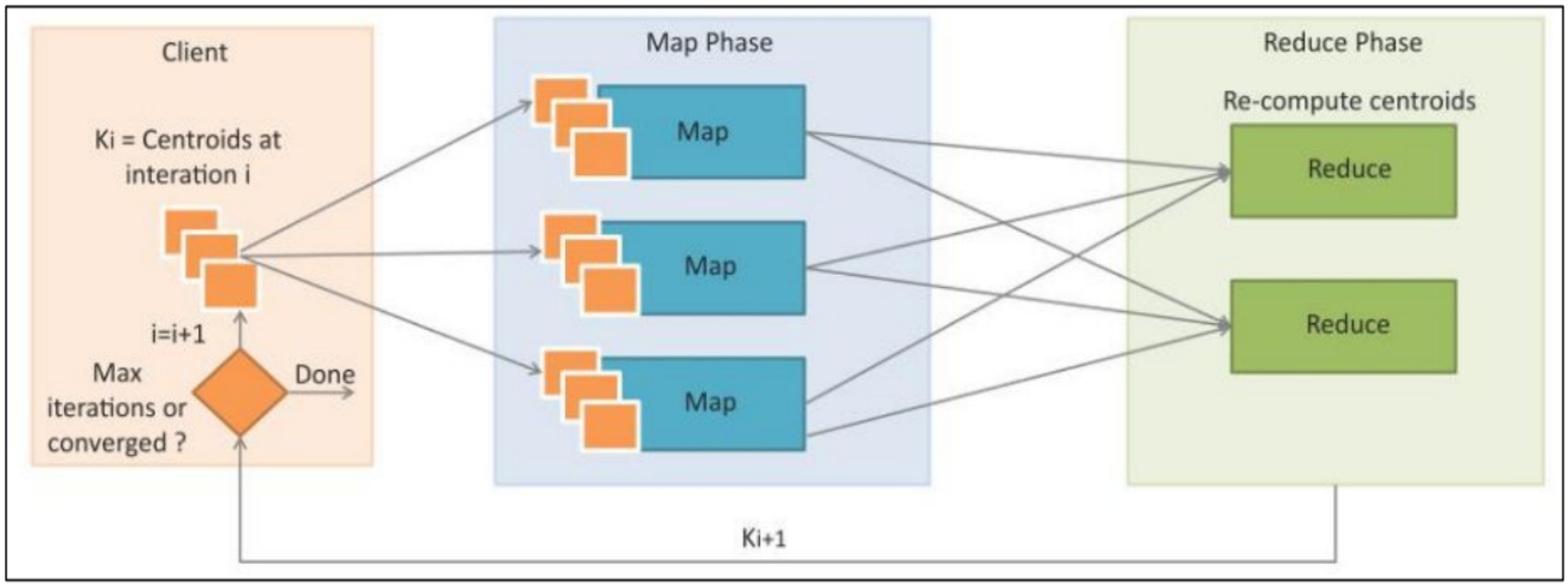
- Characteristics of big data:
  - **Velocity (Speed):** Velocity of data refers to how fast the data is generated. Modern IT, industrial and other systems are generating data at increasingly higher speeds generating big data.
  - **Variety (Complexity):** Variety refers to the forms of the data. Big data comes in different forms such as structured or unstructured data, including text data, image, audio, video and sensor data.

Clustering Big Data

- Clustering is the process of grouping similar data items together such that data items that are more similar to each other (with respect to some similarity criteria) than other data items are put in one cluster.
- Clustering big data is of much interest, and happens in applications such as: Clustering social network data to find a group of similar users, Clustering electronic health record (EHR) data to find similar patients, Clustering sensor data to group similar or related faults in a machine, etc.
- Clustering is achieved by clustering algorithms that belong to a broad category algorithms called unsupervised machine learning.
- Unsupervised machine learning algorithms find the patterns and hidden structure in data for which no training data is available.

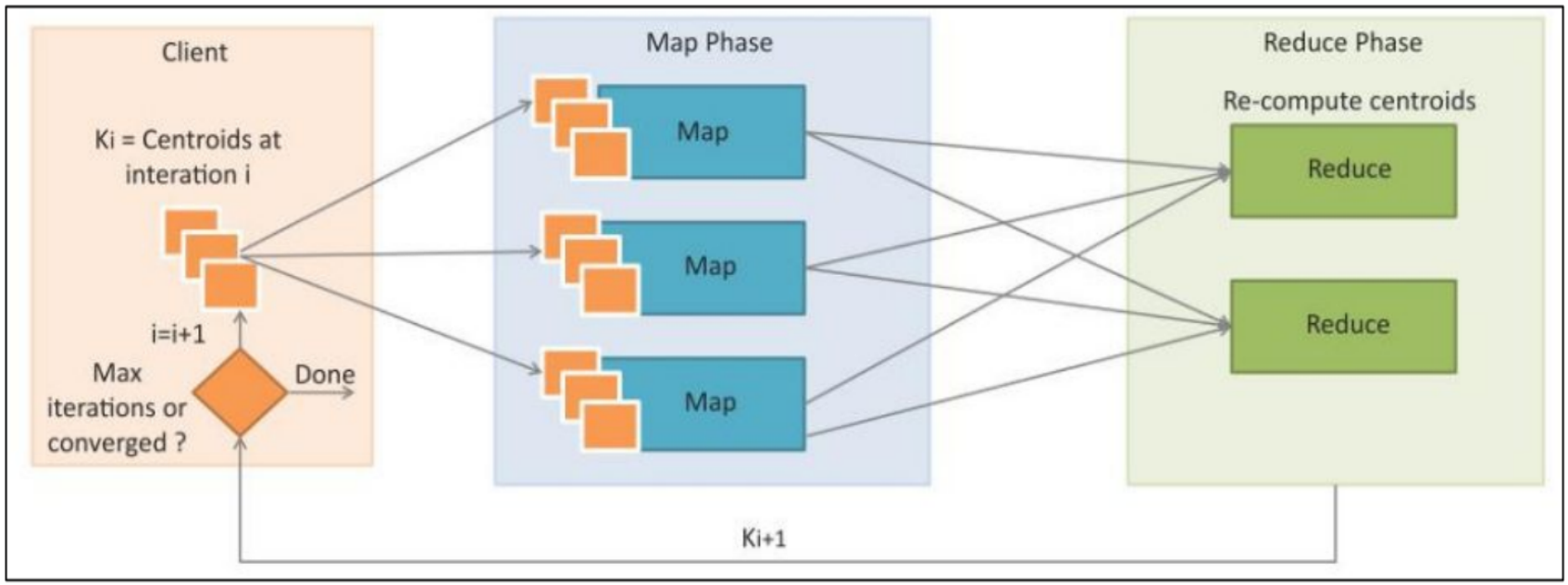
k-means with MapReduce

- The data to be clustered is distributed on a distributed file system such as HDFS and split into blocks which are replicated across different nodes in the cluster.
- Clustering begins with an initial set of centroids. The client program controls the clustering process.
- In the Map phase, the distances between the data samples and centroids are calculated and each sample is assigned to the nearest centroid.



k-means with MapReduce

- In the Reduce phase, the centroids are recomputed using the mean of all the points in each cluster.
- The new centroids are then fed back to the client which checks whether convergence is reached or maximum number of iterations are completed.



Classification of Big Data

- Classification is the process of categorizing objects into predefined categories.
- Classification is achieved by classification algorithms that belong to a broad category of algorithms called supervised machine learning.
- Supervised learning involves inferring a model from a set of input data and known responses to the data (training data) and then using the inferred model to predict responses to new data.
- **Binary classification:** Binary classification involves categorizing the data into two categories. For example, classifying the sentiment of a news article into positive or negative, etc.
- **Multi-class classification:** Multi-class classification involves more than two classes into which the data is categorized. For example, gene expression classification problem involves multiple classes.

Performance of Classification Algorithms

- **Precision:** Precision is the fraction of objects that are classified correctly.

$$Precision = \frac{TruePositive}{(TruePositive + FalsePositive)}$$

- **Recall:** Recall is the fraction of objects belonging to a category that are classified correctly.

$$Recall = \frac{TruePositive}{(TruePositive + FalseNegative)}$$

- **Accuracy:**

$$Accuracy = \frac{(TruePositive + TrueNegative)}{(TruePositive + TrueNegative + FalsePositive + FalseNegative)}$$

The End